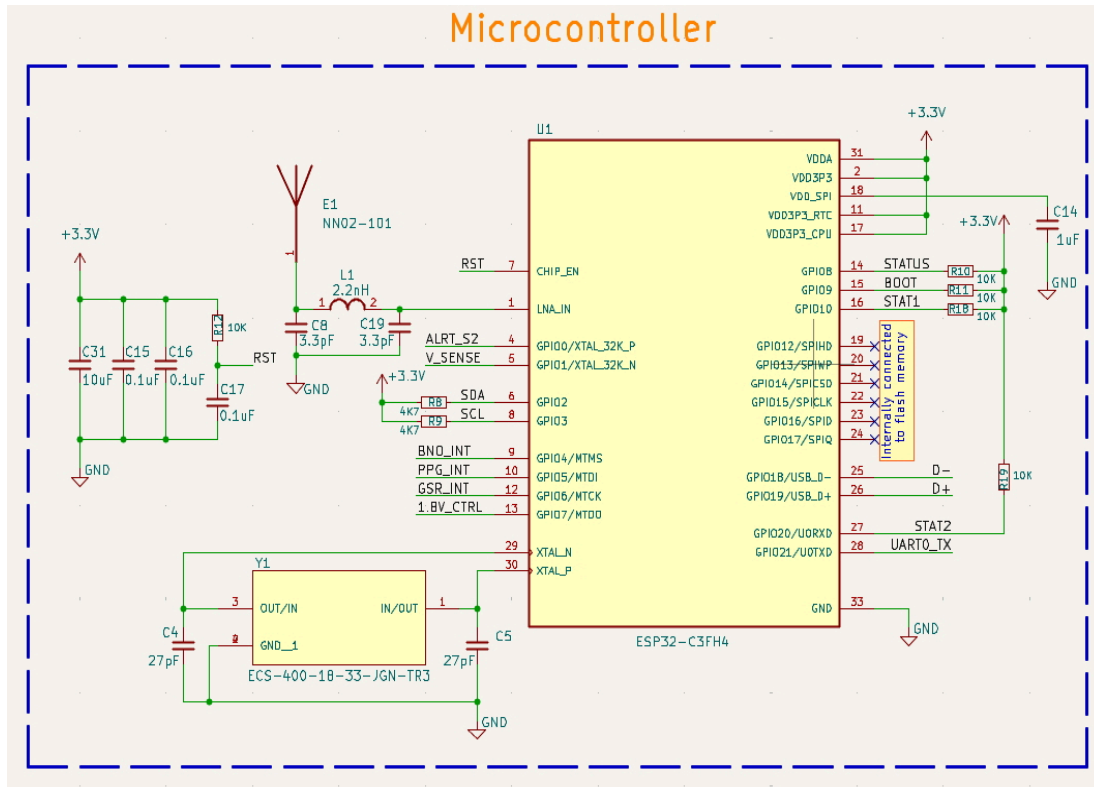


Design Review

Microcontroller ESP32-C3FH4

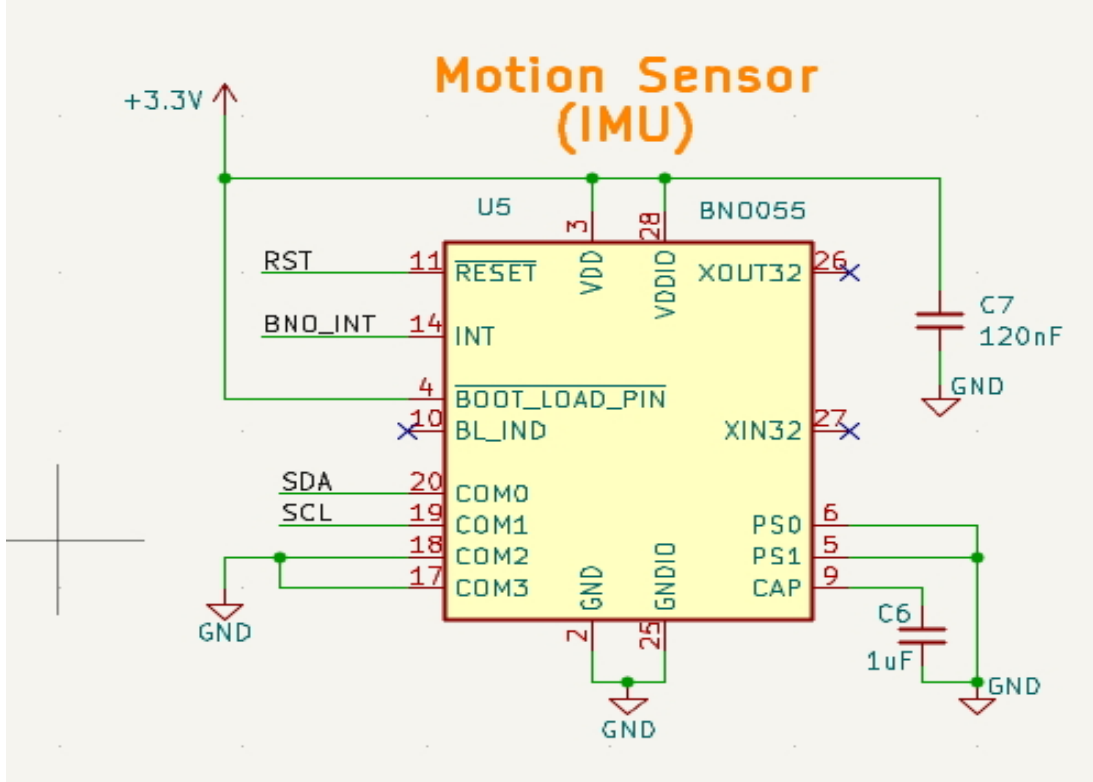


GPIO8, GPIO9, GPIO10 and GPIO20 are pulled up to 3.3V with 10K resistors. These pins are pulled up because they are ESP32-C3 strapping / boot-related pins and want them to have defined logic levels during reset/boot.

Specifically:

- GPIO8 → STATUS
- GPIO9 → BOOT
- GPIO10 → STAT1
- GPIO20 → STAT2 / U0RXD

BNO055 IMU



From the shown BNO055 schematic, your connections look mostly correct for I²C mode.

BOOT_LOAD_PIN connected to +3.3 V

This is okay for normal operation.

The BNO055 BOOT_LOAD_PIN is used to enter bootloader/programming mode. For normal use, it should be held high.

However, I would usually recommend using a pull-up resistor, for example:

COM2 and COM3 connected to GND

Yes, this is correct for I²C operation.

COM0 → SDA
COM1 → SCL
COM2 → GND
COM3 → GND

Make sure your SDA and SCL lines have pull-up resistors somewhere on the bus, typically 4.7 kΩ to 3.3 V. They may already exist elsewhere in your schematic.

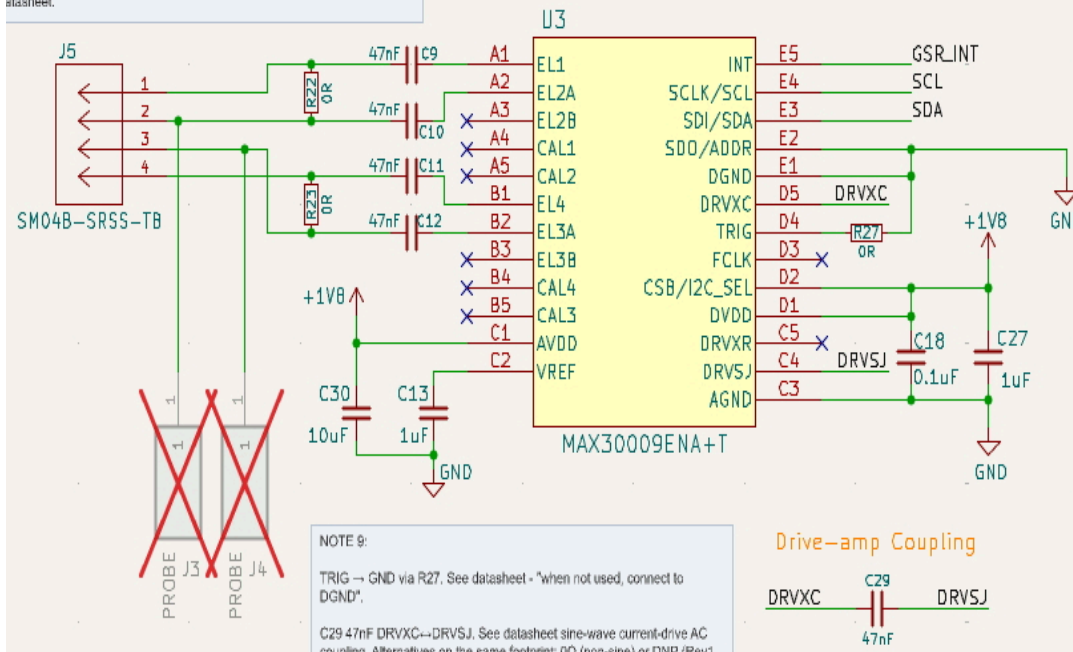
PS0 and PS1 connected to GND is okay.

Also try to have a 1 μF or 4.7 μF bulk capacitor near the BNO055 power pins.

Galvanic Skin Response

22/R23 0Ω = config links: FITTED = proven bipolar with 2-wire leads.
 REMOVE BOTH for true tetrapolar, or for Kelvin bonding at the electrode
 nd. J3 & J4 are possible board mount probes.
 9-C12 (47nF) required AC-coupling for GSR/EDA per MAX30009
 datasheet.

Electrodermal Sensor (EDA)



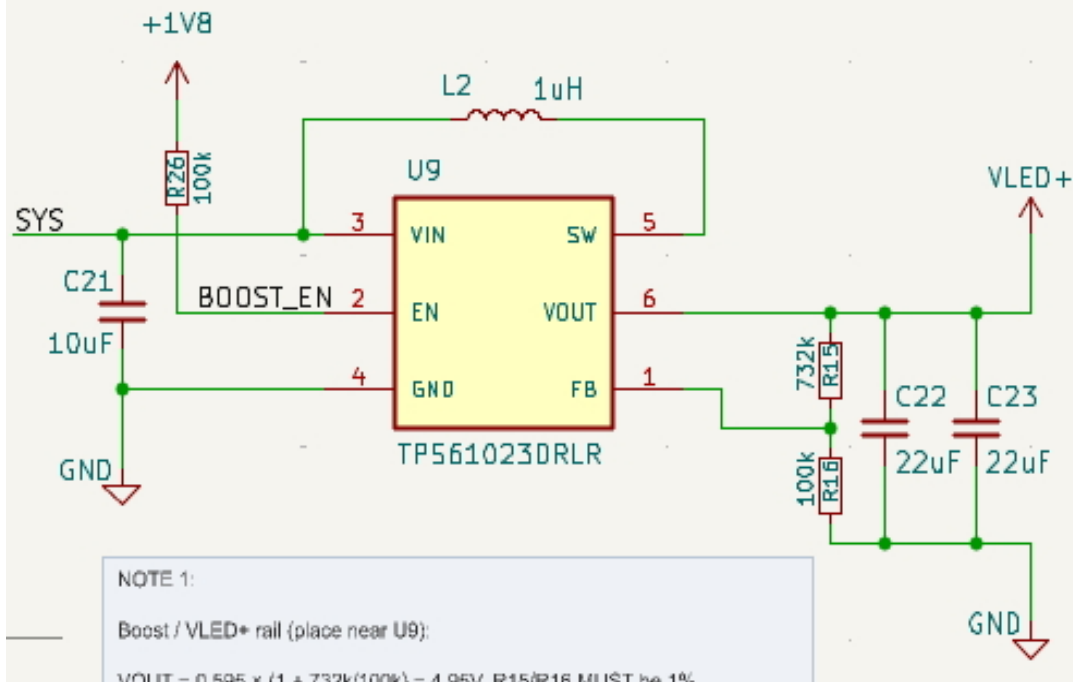
GND / VSS: Connected to Ground. Ensure these traces are as short as possible to the ground plane to maintain signal integrity, which is critical for sensitive biopotential measurements like GSR.

- RST (Pin 5): The 10 kΩ pull-up to 3.3 V is correct. The reset pin is active-low, so you must keep it high for normal operation.
- INT (Pin 6): The 10 kΩ pull-up is correct. The interrupt pin on the MAX30009 is typically open-drain, which requires this pull-up resistor to function properly when reporting events to the MCU.
- CSB (Pin 9): The 10 kΩ pull-up to 3.3 V is correct. For I²C mode, the Chip Select Bar (CSB) pin must be pulled high.
- SCL (Pin 7) & SDA (Pin 8): Both have 4.7 kΩ pull-up resistors. This is the standard recommendation for I²C buses to ensure reliable rise times at standard clock speeds (100–400 kHz).

Since this is a high-precision analog chip, ensure that the traces for GSR_A, GSR_B, DRV_A, and DRV_B are kept away from noisy digital signals (like the I²C lines or high-speed clock traces) to minimize artifacts in your skin conductance readings. Also, verify that you have a 0.1 μF or 1 μF ceramic decoupling capacitor placed as physically close as possible to the VDD/VDDIO pins.

Boost Converter LED

Boost Converter



If SYS is from a battery or noisy rail, adding another 0.1 μ F close to VIN is also helpful.

boost converter topology is basically correct.

Main checks/corrections:

Current output voltage is about 4.16 V, not 5 V.

If you want 5 V, change the upper feedback resistor to about 900k with 100k lower resistor.

Add a pull-down on EN if you want the boost off by default.

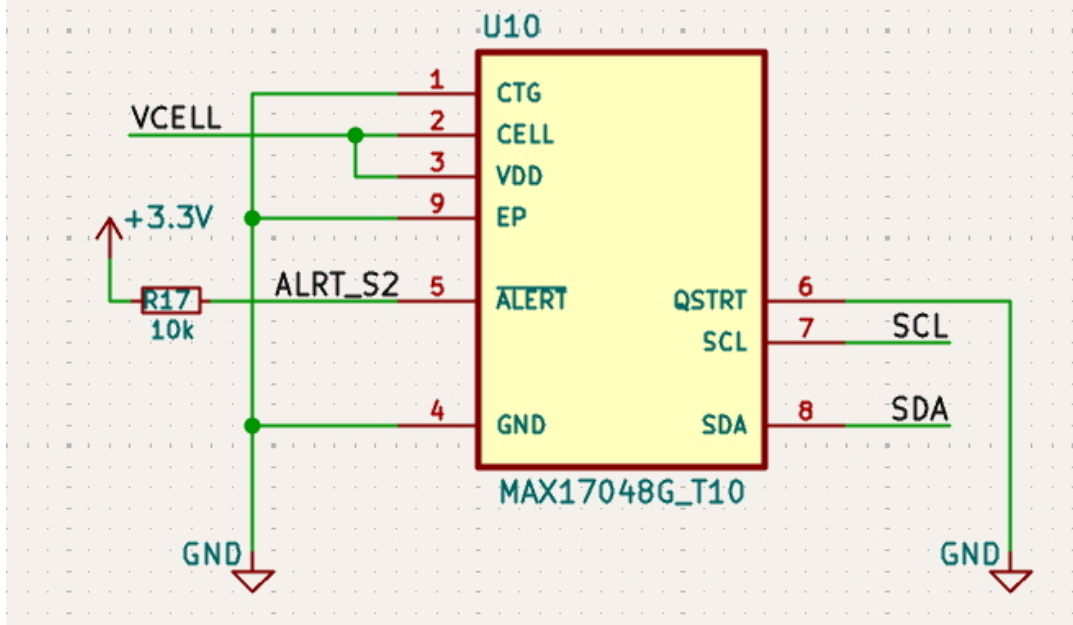
Make sure the 1 μ H inductor has enough saturation current.

Keep the SW node small and the FB trace away from switching noise.

This is a voltage boost, so LEDs still need proper current limiting.

LiPo Fuel Gauge

LiPo Fuel Gauge



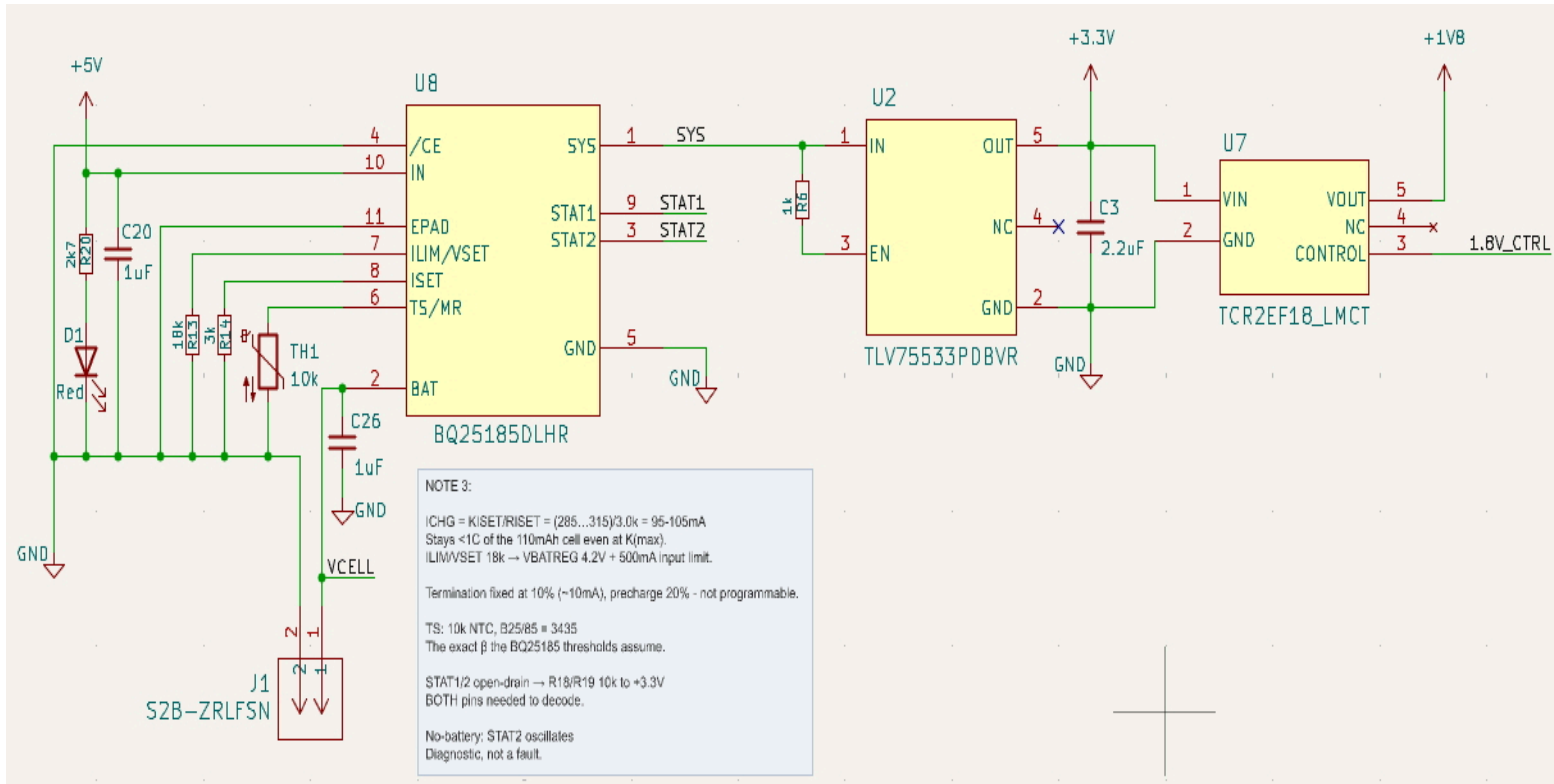
Here VCELL cannot be connected to the VDD. VDD could be connected with 3.3V regulator output.

Need 2 capacitors as below:

VDD $\rightarrow 0.1 \mu\text{F} \rightarrow \text{GND}$

VCELL $\rightarrow 0.1 \mu\text{F} \rightarrow \text{GND}$

USB Power Rail



Lipo battery charger and other power rails are correct.